Lorentz center

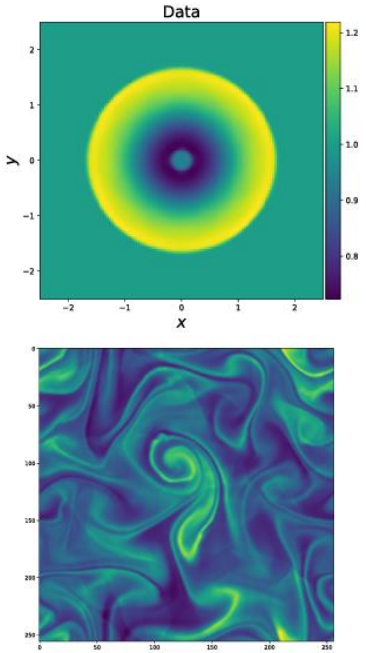
Benchmarking for Scientific Machine Learning in Subsurface Geoscience

20 - 24 April 2026, Leiden, the Netherlands **@omega**

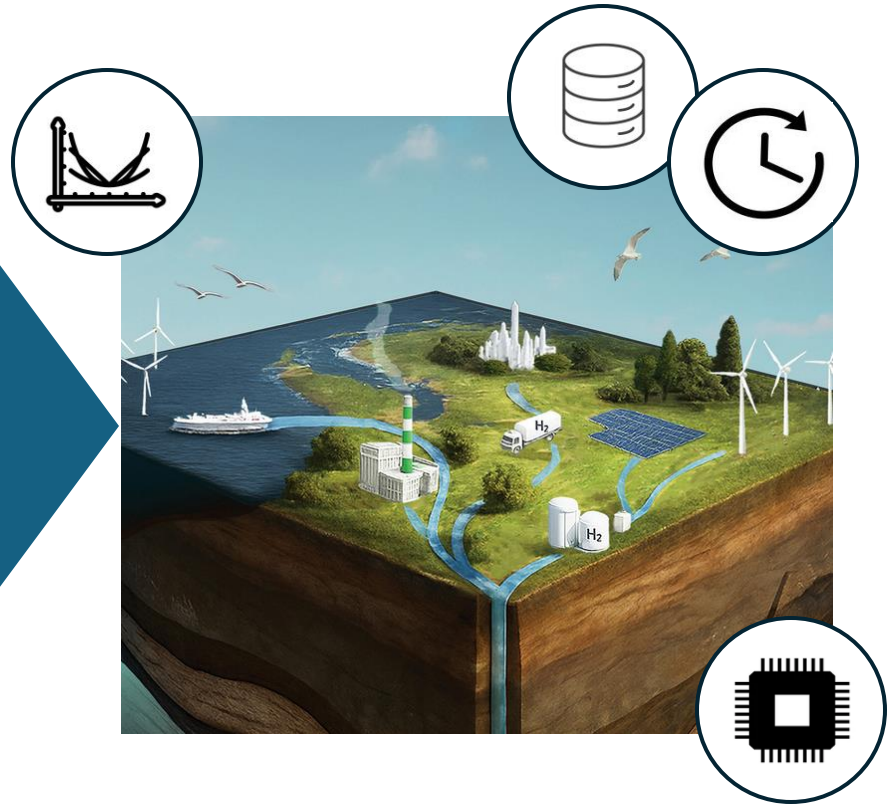
Scientific Organizers

- Alexander Heinlein, Delft University of Technology
- Julia Pelzer, University of Stuttgart
- Miriam Schulte, University of Stuttgart & University of Bergen
- Hadi Hajibeygi, Delft University of Technology
- Inga Berre, University of Bergen

Motivation



Stepwise Benchmarking

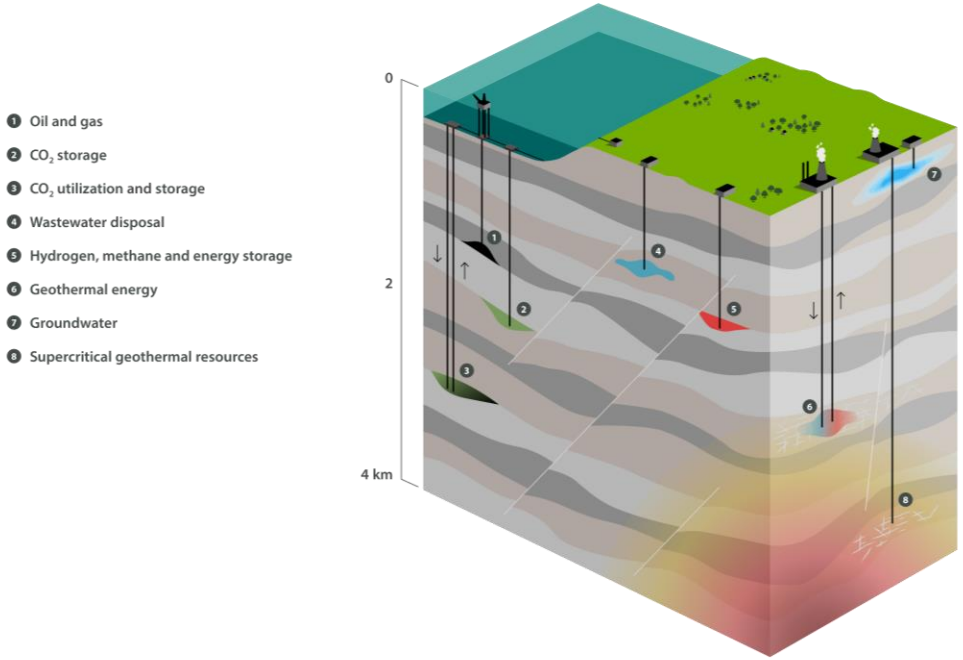


[TakamotoEtAl22]

[vanRooijenHajibeygi25]

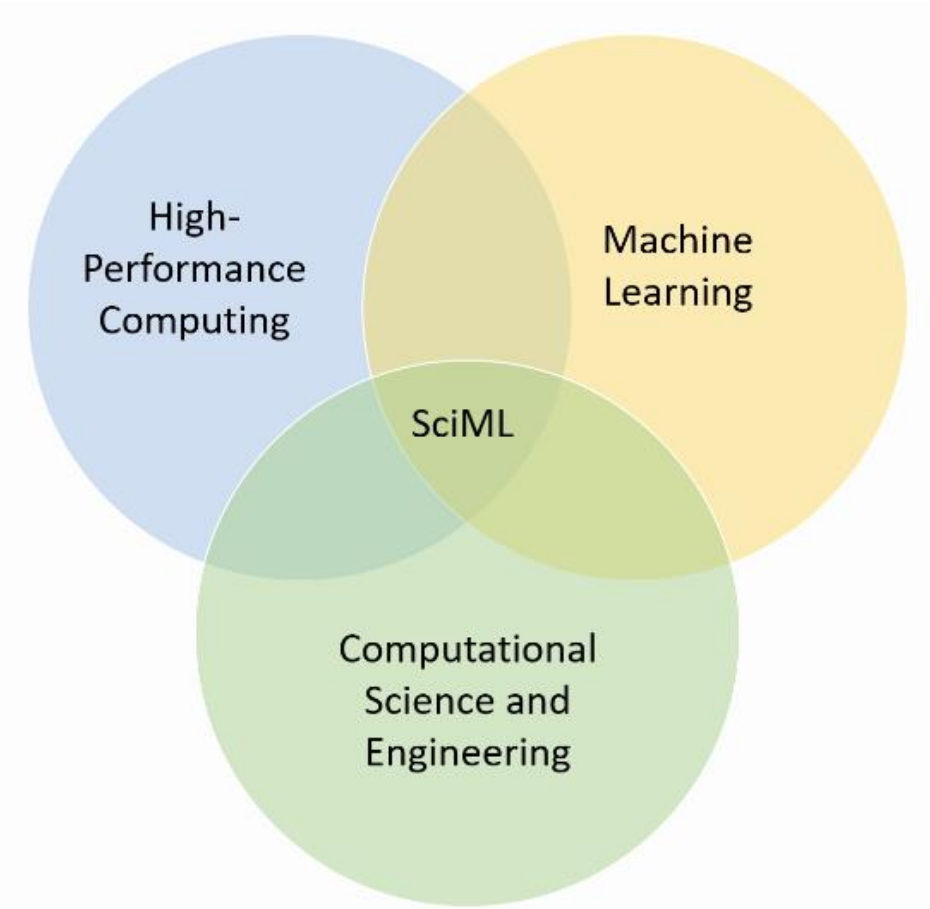
Topical Overview

Subsurface Geosciences



[Inga Berre]

Scientific Machine Learning



https://ems-tag-sciml.github.io/

Time Schedule

	Monday	Tuesday	Wednesday	Thursday	Friday
09:00-12:00	preparations / help with setup; regist-ration	contributed talks	reporting from interactive	reporting from interactive	workshop report
			coffee break	coffee break	coffee break
		coffee break	contributed talks	interactive	panel discussion
		orga and start			
	opening Lorentz	interactive	interactive		*closing*
12:00-13:00	lunch break	lunch break	lunch break	lunch break	lunch break
13:00-18:00	*opening*				
	talk: application	interactive session	interactive session	interactive session	
	talk: benchmark				
	coffee break				
	talks: benchmarks	coffee break	coffee break	coffee break	
				reporting	
		poster session	interactive session	next-steps discussion	
	welcome reception		*walk/ride to*		
18:00-			conference dinner		

Time Schedule - Rest of Today

13.00-13.15	Organizers	Welcome and Opening
13.15-14.15	Denis Voskov	Role of Machine Learning in Subsurface Modelling
14.15-14.45	Julia Pelzer	Stepwise Benchmarking in the Data Scarce Setting of Groundwater Heat Plume Modeling
14.45-15.15		*coffee break*
15.15-15.40	Bernd Flemisch	The FluidFlower and SPE11 benchmarking studies
15.40-16.00	Antoine Lechevallier	ML4GCS Project 2 - History matching SPE11B data
16.00-16.30	Jakub Both	Objective comparison and hierarchical clustering of multiphase flow data
16.30-17.00	Retief Lubbe	Geophysical granular flow benchmarks over frictional and bumpy surfaces
17.00-19.00		*welcome reception @Lorentz center*

Sources

- Real-world example: van Rooijen & Hajibeygi, ACS Energy & Fuels, 39, 41, 19600–19613 (2025)
- Piktograms:
  - https://www.shutterstock.com/de/image-vector/time-icon-black-clock-symbol-flat-1277165743
  - https://www.freepik.com/free-photos-vectors/pictogram-database
  - https://www.shutterstock.com/de/image-vector/chip-icons-set-microchip-integrated-circuit-2318538885
  - https://www.vecteezy.com/vector-art/54190376-loss-function-chart-machine-learning-line-icon-illustration

- Benchmarks toy: Takamoto, Makoto, et al. "Pdebench: An extensive benchmark for scientific machine learning." *Advances in neural information processing systems* 35 (2022): 1596-1611.
- SciML: https://ems-tag-sciml.github.io/
- Geosciences: Inga Berre